

Human Learning VS Machine Learning

Chutisant Kerdvibulvech, Ph.D.
Assistant Professor

ผศ.ดร.ชุตินันต์ เกดวิบูลย์เวช

Learning and Development in the Workplace

OVERVIEW

- Speaker's Background
- Machine Learning
- Human Learning: What Human can Beat Machine
- Related-recent technologies: Big Data, Computer Vision, Augmented Reality

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MYSELF

- Received PhD in Computer Engineering from Keio University, Japan (2009)
- Assistant Professor at GSCM, NIDA (2015-Now)
- Head of Department of Information and Communication Technology, Rangsit University (2009-2014)

EDUCATION



www.chula.ac.th/

- B.E. (Hons) in Computer Engineering
Chulalongkorn University



www.keio.ac.jp

- MSc. / Ph.D. in Engineering, Department of Information and Computer Science
Keio University, the oldest university in Japan

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ACADEMIC WORK EXPERIENCE

- Invited Lecturer and Research Committee in many universities in Thailand, such as
 - Chulalongkorn University
 - Mahidol University
 - Sripatum University
 - Kasetsart University
 - Panyapiwat Institute of Management
- Technical Committees for many academic international conferences and journals (Scopus and Scimago)

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MEDIA WORK EXPERIENCE

Newspapers

- IT Columnist for Dailynews newspaper, every Wednesday (page 23)



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MACHINE LEARNING

CHATBOT



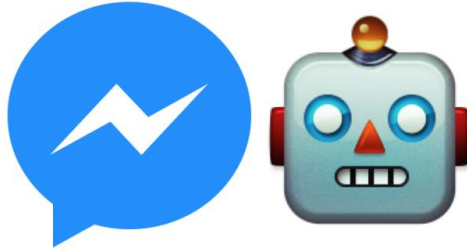
<http://knowledge.wharton.upenn.edu/article/rise-chatbots-time-embrace/>

“A chatbot (also known as a talkbot, chatterbot, Bot, chatterbox, Artificial Conversational Entity) is a computer program which conducts a conversation via auditory or textual methods.”

<https://en.wikipedia.org/wiki/Chatbot>

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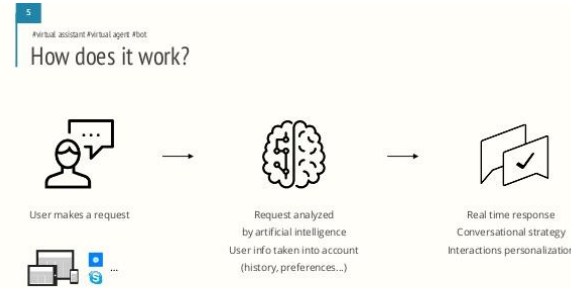
CHATBOT = CHAT + ROBOT



<http://www.garethjames.net/complete-guide-chatbots/>

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CHATBOT: HOW DOES IT WORK?



<http://www.garethjames.net/complete-guide-chatbots/>

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MACHINE LEARNING

- “Machine learning is the subfield of computer science that gives computers the ability to learn without being explicitly programmed” (Arthur Samuel, 1959).
- “Evolved from the study of pattern recognition and computational learning theory in artificial intelligence,[2] machine learning explores the study and construction of algorithms that can learn from and make predictions on data” Ron Kohavi; Foster Provost (1998). "Glossary of terms". *Machine Learning*. 30: 271-274.

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DEEP LEARNING

- Deep Machine Learning
- “A branch of machine learning based on a set of algorithms that attempt to model high level abstractions in data”
- Various deep learning architectures: **computer vision**, automatic speech recognition, and natural language processing

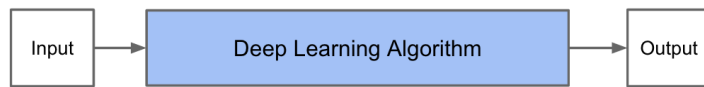
Bengio, Y.; et al. "Representation Learning: A Review and New Perspectives". *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 35 (8): 1798-1828

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MACHINE LEARNING VS DEEP LEARNING



Traditional Machine Learning Flow



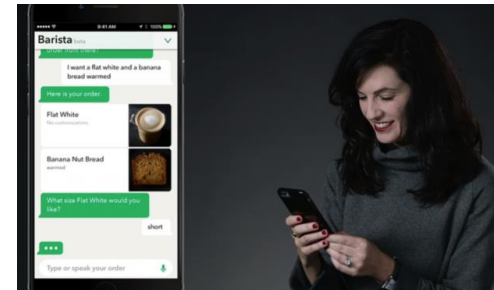
Deep Learning Flow

<http://adilmoujahid.com/posts/2016/06/introduction-deep-learning-python-caffe/>

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STARBUCKS UNVEILS CHATBOT APP (1)

- A voice-activated chatbot, called MyStarbucks Barista



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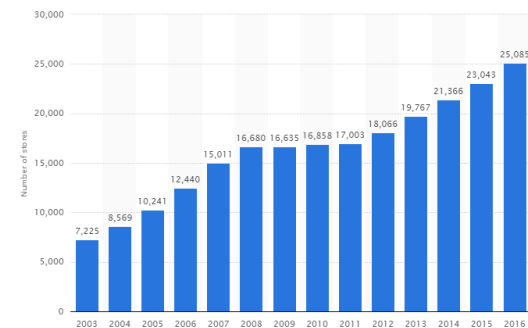
STARBUCKS UNVEILS CHATBOT APP (2)

- Let customers place orders by speaking, "I want a latte." The features can also be used to pay for orders.
- MyStarbucks Barista is only available to 1,000 people before a nationwide release this summer.

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STARBUCKS UNVEILS CHATBOT APP (3)

Number of Starbucks stores worldwide from 2003 to 2016



<https://www.statista.com/statistics/266465/number-of-starbucks-stores-worldwide/>

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MELODY (1)



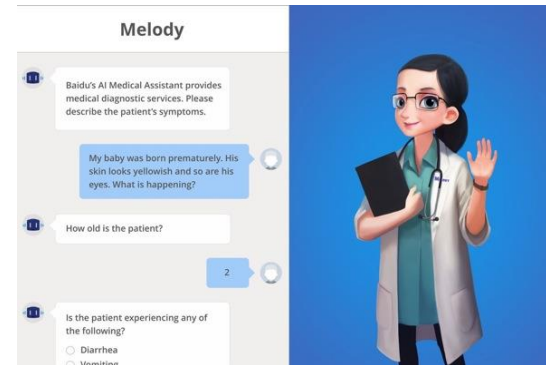
<http://www.cnbc.com/2016/10/11/chinese-baidu-unveils-ai-health-chatbot-for-patients-and-doctors.html>

- Baidu, China's biggest search engine
- Launches medical chatbot to help Chinese doctors diagnose patients

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- Built into the company's iOS and Android Baidu Doctor app, which launched in China in 2015

MELODY (2)



<http://venturebeat.com/2016/10/11/baidu-launches-medical-assistant-chatbot-to-help-doctors-collect-patient-information/>

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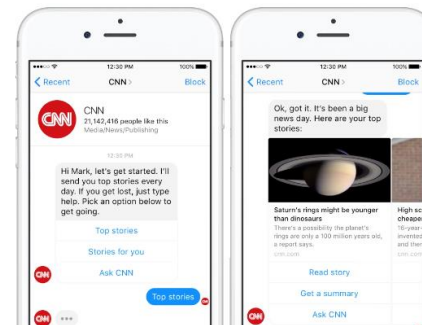
BOTS FOR MESSENGER: CNN (1)

- “Facebook announced at its F8 conference that it has created a platform for companies to develop the robots that mimic human conversation to run within its Messenger app”
- “Bots for Messenger enables users to get everything from weather updates to shopping notifications and personalized news as well as message companies and publishers as they do their friends.”

<http://www.brandchannel.com/2016/04/14/brand-chatbots-041416/>

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BOTS FOR MESSENGER: CNN (2)

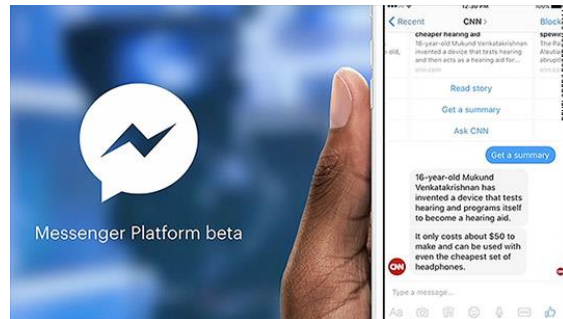


<http://www.brandchannel.com/2016/04/14/brand-chatbots-041416/>

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- “Launch partners include shopping app Spring, HP and CNN.”

BOTS FOR MESSENGER: CNN (3)



<http://money.cnn.com/2016/04/12/technology/facebook-messenger-bots/>

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HUMAN LEARNING

WHAT HUMAN CAN BEAT MACHINE? (1)

- Sensing emotion—anxiety, stress, sadness, frustration, happiness, excitement
- Loving and caring for children and the elderly
- Creating art / Finding beauty in everyday life

<https://medium.com/women-in-product/25-things-a-human-can-beat-a-machine-at-bd00e2873c9f>

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WHAT HUMAN CAN BEAT MACHINE? (2)

- Finding common ground
- Negotiating a deal that benefits both sides
- Imagination / Creativity
- No routine work

<https://medium.com/women-in-product/25-things-a-human-can-beat-a-machine-at-bd00e2873c9f>

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RELATED- RECENT TECHNOLOGIES

FUTURE OF INNOVATION?



www.3dsg.com.au

- AI (Artificial Intelligence) / Deep Learning
- Big Data / Thick Data
- Computer Vision / Augmented Reality
 - 3D-Display / 3D Television
 - MR (Mixed Reality) / AR (Augmented Reality) / VR (Virtual Reality)

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BIG DATA

- “A term for data sets that are so large or complex that traditional data processing application software is inadequate to deal with them”

“The World’s Technological Capacity to Store, Communicate, and Compute Information”. MartinHilbert.net.

- Capturing data, data storage, data analysis, search, sharing, transfer, visualization, querying, updating and information privacy

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BIG DATA IS ALWAYS CORRECT?

- Machine is always correct?
- To make the right business decisions and thrive in the unknown
- Case study of Nokia
- Dr. Tricia Wang: The human insights missing from big data
 - PhD in Sociology from UC San Diego
 - Holds position at Data & Society, Harvard University’s Berkman Klein Center for Internet Studies
 - A global tech ethnographer

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CASE STUDY: THE FALL OF NOKIA (1)

Why do so many companies make bad decisions, even with access to unprecedented amounts of data?

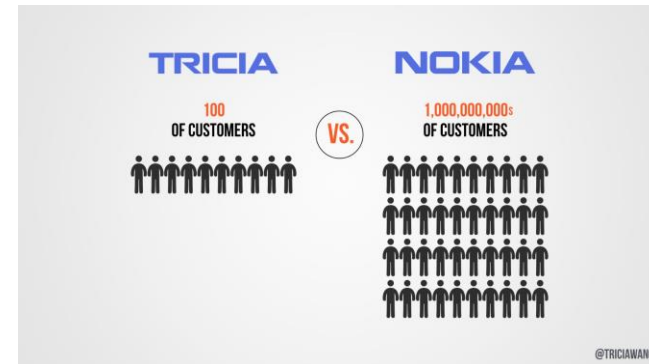


https://www.ted.com/talks/tricia_wang_the_human_insights_missing_from_big_data

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CASE STUDY: THE FALL OF NOKIA (2)

Human VS Machine



<https://medium.com/ethnography-matters/why-big-data-needs-thick-data-b4b3e75e3d7>

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THICK DATA

- Qualitative information that provides insights into the everyday emotional lives of consumers
- Goes beyond big data to explain why consumers have certain preferences, the reasons they behave the way they do, why certain trends stick and so on

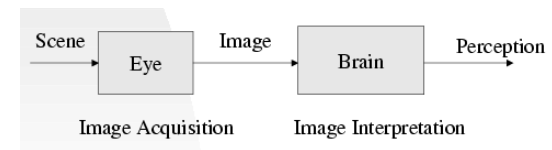
<http://bigfishpr.com/the-power-of-thick-data/>

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VISION

- Vision is the process of discovering what is present in the world and where it is by looking.



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COMPUTER VISION

- “science and technology of machines that see”
- As a scientific discipline, computer vision is concerned with the theory behind artificial systems that extract information from images
- The image data can take many forms, such as video sequences, views from multiple cameras, or multi-dimensional data from a medical scanner

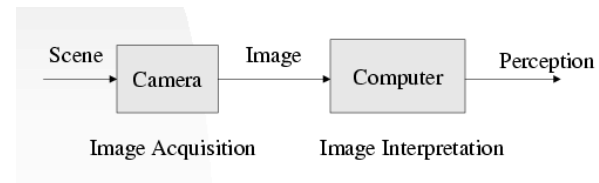
By Dana H. Ballard, Computer Vision.

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COMPUTER VISION

- Computer Vision is the study of analysis of pictures and videos in order to achieve results similar to those as by men.



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3D DISPLAY / 3D TELEVISION

3D DISPLAY / 3D TELEVISION



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3D DISPLAY / 3D TELEVISION

A free-floating image created by firing lasers into thin air



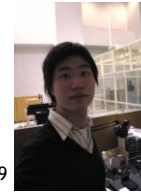
<http://www.afp.com/>

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3D DISPLAY / 3D TELEVISION

2014 October

- 2008-Current
- Hideki Kimura
- National Museum of Emerging Science and Innovation (Miraikan), Tokyo



2009

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<http://www.afp.com/>

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3D DISPLAY / 3D TELEVISION



<http://www.afp.com/>

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3D DISPLAY / 3D TELEVISION

- Miraikan is located in Odaiba, Tokyo



www.whitehouse.gov



www.japan-guide.com

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NEW VIEWPOINT ANALYSIS

Prof. Takeo Kanade,
Carnegie Mellon University (CMU), United States

SUPER BOWL

- Eye Vision
- Involves shooting multiple video images of a dynamic event
- The video streams combined by computer and the resulting images reach viewers as if they are flying through the scenes they see



Eye Vision from Prof.
Takeo Kanade, CMU

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THE MATRIX



www.imdb.com

- “Virtualized Reality”
- People will be able to customize the perspective from which they watch, for example, from that of a particular object.

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AUGMENTED REALITY

WHAT IS AUGMENTED REALITY (AR)?



- A combination of a real scene viewed by a user and a virtual scene generated by a computer that augments the scene with additional information.
- Generates a composite view for the user.

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AUGMENTED REALITY APPLICATIONS

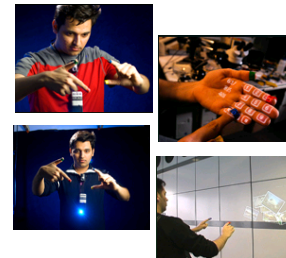
SIXTH SENSE

- A wearable gestural interface that augments the physical world around us with digital information
- Let us use natural hand gestures to interact with that information
- The SixthSense prototype is comprised of a pocket projector, a mirror and a camera.

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SIXTH SENSE IN 21ST CENTURY



TED
TALKS

Dr. Pattie Maes,
Massachusetts Institute of Technology
(MIT), United States

Broadcast in TED

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DETAIL OF SIXTH SENSE

- Sixth Sense allows people to use the internet without a screen or a keyboard
- It can turn any surface into a touch screen
- It is made with store-bought components
- *“The software program processes the video stream data captured by the camera and tracks the locations of the colored markers [that you wear on your fingers] using simple computer-vision techniques”..from MIT Media Lab*

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AUGMENTED REALITY GUITAR

AUGMENTED REALITY GUITAR



Dr. Chutisant Kerdvibulvech,
Keio University, Japan

Broadcast in Discovery Channel



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QUESTIONS?

Asst. Prof. Dr. Chutisant Kerdvibulvech
Graduate School of Communication Arts and Management Innovation,
National Institute of Development Administration (NIDA)
chutisant.ker@nida.ac.th
<https://www.facebook.com/chutisant.kerdvibulvech>

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